

TODDLER'S LEARNING GAME

**A System Analysis Design
Presented to the Faculty of
College of Computer Science and Information Technology
Golden West Colleges
Alaminos City, Pangasinan**

**In Partial Fulfillment
Of the Requirements in the Degree Program
Bachelor of Science in Information Technology**

By:

Duguinay, Generoso III, C.

Ganaden, Cherry Ann D.

Ballares, Markshane G.

Ballado, Rogelio R.

Bernal, Jovelyn U.

Corbillon, Aj A.

**Mr. John Vianney V. Manuel
Adviser**

December 2024

GOLDEN WEST COLLEGES

San Jose Drive, Poblacion, Alaminos City, Pangasinan

(075) 552 - 7382 goldenwest.colleges@yahoo.com.ph

This System Analysis and Design Project entitled, "Toddler's Learning Game" prepared by "Generoso C. Duguinay, Cherry Ann D. Ganaden, Markshane G. Ballares, Rogelio R. Ballado, Jovelyn U. Bernal, and Aj A. Corbillon" in partial fulfillment of the requirements in Bachelor of Science in Information Technology have been examined and is recommended for your approval and acceptance this December 16, 2024.

JOHN VIANNEY V. MANUEL
System Analysis and Design, Adviser

THE ORAL EXAMINATION COMMITTEE

Oral Examination Grade

KAIROS JIL I. MOLINA
Chairman

LUIELYN B. RAGUINDIN
Member

PAULO M. BALGUA
Member

APPROVED and ACCEPTED
in partial fulfillment of the requirements in
Bachelor of Science in Information Technology.

KAIROS JIL I. MOLINA
Dean, CSIT Department

Abstract

Duguinay, Generoso III C., Ganaden, Cherry Ann D., Ballares Markshane G., Ballado, Rogelio R., Bernal, Jovelyn U., Corbillon, Aj A. "Toddlers Learning Game", Bachelor of Science in Information Technology, College of Information Technology Education, Alaminos City, Pangasinan, Philippines.

Adviser: John Vianney V. Manuel

This study focuses on the mental development of the toddlers (3-6 y/o). Educational games are often used in a teaching approach known as game-based learning. The goal of this study is to educate and teach the toddlers and promote simple vocabulary.

Objectives of the study were stated to direct the emphasis of the work. Rapid Application Development (RAD) is a software development technique that was used to design and develop this application. The developers used quantitative methods to get reliable results and a clear research goal.

This study will help the toddlers to enhance their brain development at an early age.

Keywords: Learning games for toddlers using the Toddler's Learning ga

and lost hope that we might not be able to finish it, but because of them we encouraged ourselves to do our best no matter what.

We also want to give appreciation to those people we met while doing the survey questions. Thank you for allowing us to talk and have a survey with you. And finally, we would like to thank God for guiding us through all the difficult times and keeping us away from harm. We will always put our faith in you for the rest of our lives.

Generoso C. Duguinay III

Cherry Ann D. Ganaden

Markshane G. Ballares

Rogelio R. Ballado

Jovelyn U. Bernal

Aj A. Corbillon

Acknowledgement

We would like to acknowledge and give the warmest thanks to the most patient person who guided us to overcome this study, thanks to our beloved adviser Mr. John Vianney Manuel, for guiding us in this study for this semester. Despite his busy schedule and even if he is very tired sometimes because of us, he always insisted on teaching, guiding and correcting us multiple times. We would not have been able to complete our thesis without the help of our adviser. Thank you so much for being a hands-on adviser to us and always reminding us that it's okay to be tired, just don't give up. We would also like to mention Ms. Luielyn Raguindin for advising us in making our documents. Thank you also for your time and patience.

We also want to cherish our classmates and friends for being there because somehow, they helped us and made us laugh and made things less daunting.

We also want to express our thankfulness to our boyfriends and girlfriends for always encouraging and helping us to finish our thesis writings and systems. The work is faster with your help and presence.

We would also like to give a special credit to our family members for their assistance and support throughout our thesis journey. During the thesis writing process, we almost felt weak

Dedication

I dedicated this system to my parents because they did not expect that we could do this system and they trusted in the ability that we have, and also to the Lord because we prayed before we did and started our system.

Jovelyn Bernal

I dedicated this system first to God because he gave me strength to finish it and of course to my family and friends who pushed us to get through it. Thank you.

Cherry Ann Ganaden

I dedicated this system to my parents Alvin Corbillon and to my mom Gemma Alvarez because they did not expect that we could do this system and they trusted in the ability that we have, and also to the Lord because we prayed before we did and started our system.

Aj Corbillon

This research is dedicated to my family, whose unwavering support and encouragement have been my guiding light throughout this journey. To my parents, for their endless love and belief in my potential. And to my friends, for their understanding and patience during the countless hours spent on this work.

Generoso Duguinay III

I dedicated this system to my parents Elisa Mejos and to my father Rogelio Ericson Ballado because they did not expect that we could do this system and they trusted in the ability that we have, and also to the Lord because we prayed before we did and started our system.

Rogelio Ballado

I dedicated this system to my parents because they are my strength and they trusted in my ability that I can do this, and also to the Lord because he is the reason why I'm here today.

Markshane Ballares

TABLE OF CONTENTS

Title Page	i
Approval Letter	ii
Abstract	iii
Acknowledgement	iv
Dedication	vi
Table Of Contents	viii
List of Figures	xi
List of Plates	xii

Synthesis and Research Gap	16
3 DESIGN & METHODOLOGY	17
Research Design	17
Software Developmental Model	18
The Stages of Rapid Application Development	20
Source of Data	21
Instrumentation and Data Collection	22
Tools for Data Analysis	23
4 FINDINGS & DISCUSSION	27
Flow Chart	27
Graphical Representation of Data	28
Problem Areas	33
Developed Game	33
5 SUMMARY, CONCLUSION, & RECOMMENDATION .	39
Summary of the Findings	39
Conclusions	40
Recommendations	41
BIBLIOGRAPHY	42
APPENDICES	46
APPENDIX A	46
APPENDIX B	47
APPENDIX C	48
APPENDIX D	49

APPENDIX E	50
CURRICULUM VITAE	51

List of Figures

Figure 1 Conceptual Framework	3
Figure 3.1 Rapid Development Cycle	9
Figure 4.1 Flow Chart of the Game	27
Figure 4.2 Survey Question No.1	28
Figure 4.3 Survey Question No.2	28
Figure 4.4 Survey Question No.3	29
Figure 4.5 Survey Question No.4	29
Figure 4.6 Survey Question No.5	30
Figure 4.7 Survey Question No.6	30
Figure 4.8 Survey Question No.7	31
Figure 4.9 Survey Question No.8	31
Figure 4.10 Survey Question No.9	32
Figure 4.11 Survey Question No.10	33

CHAPTER 1 INTRODUCTION

Background of the Study

Educational games are often used in a teaching approach known as game-based learning, which uses game frameworks in learning to communicate and reinforce concepts and build learners' skills in various areas. The most basic educational games definition in the classroom is that the instructor introduces a game, and the learners play the game as part of the learning process. Games and game-based learning are used because they have many benefits; especially, games and play are strongly connected to children's cognitive development and help develop many cross-curricular skills as a result.

Personal Social and Emotional Development (PSED) supports children to have a positive sense of themselves, respect for others, social skills, emotional well-being and a positive disposition to learning. Bracknell-forest council in their first few years of life, young children acquire social and emotional skills, such as regulating emotions, sharing with others and following instructions. These skills lay the foundation for developing literacy, numeracy and other cognitive abilities that are critical for success in school and life.

In this generation, some children aged 3 to 6 years old learned how to use gadgets like cellphones, iPad and sometimes laptop as well.

Toddlers know where the YouTube at, or even how to open the phones and click the games they wanted. Due to lack of guidance, some toddlers and children use gadgets too much; it takes them 1 hour or more than that playing games or watching their favorite kid's show.

To address these issues, this application called educational games or Toddler's learning games is used to help your toddlers, as well to enhance their behavior, brain development and skills. It has an important role in early childhood development.

In addition, educational game-based learning can also enhance their social, mental, and emotional development at their young age.

Conceptual Framework

The Conceptual Framework as presented in Figure 1 indicates the input variables, phases of software development and the output: The Golden West College's Toddler's Learning Game, using RAD (Rapid Application Development).

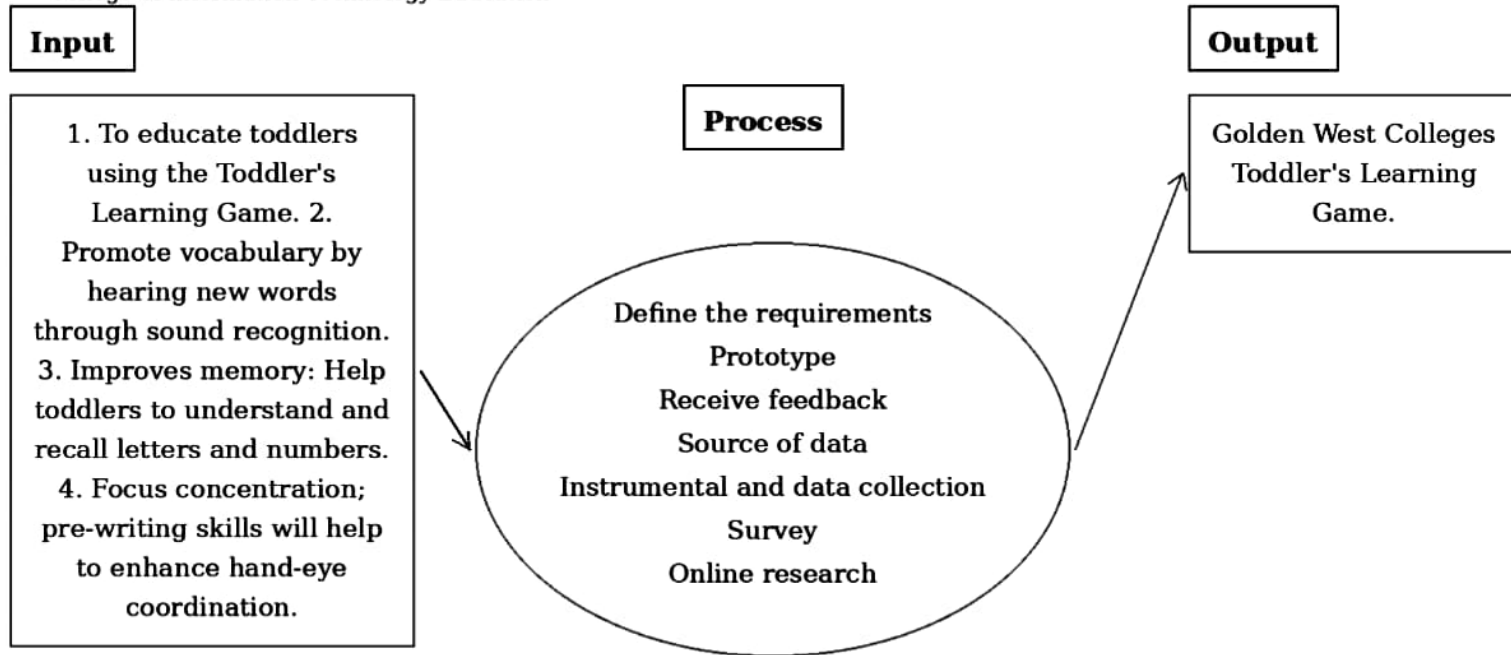


Figure 1.0 Conceptual Framework

Statement of the Problem

This study aims to address the challenges especially for toddlers by developing and honing their brain at a young age. The study seeks to:

- 1) Develop toddlers' brain and to learn new things at very young age,
- 2) Determine whether the toddlers are mentally prepared to attend school,
- 3) Evaluate the toddlers' knowledge on how well the toddlers do in activities,
- 4) Assess the mental, social, and emotional capacity of the toddlers' brain.

Objective of the Study

The goal of this game is to expand toddlers' knowledge that focuses on developmental milestones.

- To educate toddlers using the Toddler's Learning Game;
- Promote vocabulary by hearing new words through sound recognition;
- Improves memory: Help toddlers to understand and recall letters and numbers; and
- Focus and concentration: pre-writing skills will help to enhance hand-eye coordination.

Significance of the Study

The findings of this study are expected to benefit the following:

- **Toddler's (3-6 yr. old).** To test them through trial and error, find solutions to problems and recall words, sounds and pronunciation.
- **Parents and Guardian.** Adults use educational games to teach their child and toddlers to enhance early literacy and numeracy skills.
- **Daycare Center.** Can be used wherein they can use TV screen to show toddlers and children the numbers, letters, sounds, and words.

Scope and Delimitation of the Study

This study will focus on toddlers aged 3-6 years old to improve their physical, social, mental, and emotional well-being, obtaining their new skills, knowledge and behavior and helping them to learn quickly. It helps determine whether the toddler is ready for school or not.

There are limitations while using this game: a maximum of 1 hour per day, and don't force the toddlers to play when the toddlers don't want to play. This learning game is designed to help them learn in advance but with the guidance of the parents and guardian.

There are some categories contained in it such as letters, numbers, sounds, shapes, colors, animals, and vehicles.

Definition of Terms

Appropriate. Suitability or correctness of something based on context, social norms, or standards.

Assimilation. When they experience something new and try to make sense of it by relating it to something they already know.

Autonomy. Is a multifaceted concept that can be understood in different ways depending on the context.

Database. It is designed to efficiently store, retrieve, and manipulate data in a structured way.

Essential. Something that is absolutely fundamental for the existence, functioning, or success of something.

Gratification. Emotional response motivates continued interaction and reinforces learning behaviors.

Infinite. Refers to something that has no limits or end.

Meta-analysis. It provides a high-level synthesis of evidence to draw broader conclusions.

Pertinent. Something that is relevant, directly related, or significant to a particular matter or situation.

Population. Represents the entire group that is relevant to a particular question or experiment.

Preventive. Refers to actions or measures taken to stop something undesirable from happening or to reduce the risk of a negative outcome.

Prototype. A product, system, or concept that is built to test and demonstrate the core functionality, features, or design.

Receptive vocabulary. Includes the words they can recognize and comprehend when they hear them.

Significant correlations. Provide useful insights, they do not confirm causality and must be interpreted carefully to avoid misinterpretation.

Structure. Refers to the arrangement of parts or elements that make up a whole.

Theoretical framework. Provides the foundation theories and principles that guide its design and implementation.

Wide criterion. A broad or expansive standard or principle used to judge, evaluate, or measure something.



CHAPTER 2

REVIEW OF RELATED LITERATURE AND STUDIES

This chapter presents a comprehensive review of related literature and studies that served as the foundation for this research. It aims to provide an overview of existing knowledge on the topic of our game, identify research gaps, and establish a theoretical framework for the current study.

Foreign Literature

According to Alotaibi (2024), Through the use of games, this approach teaches a range of subjects and skills while promoting engagement, motivation, and enjoyment. The cognitive, social, and emotional development of young children in early childhood education may be aided by game-based learning. The purpose of this systematic review and meta-analysis is to gather the literature on the effectiveness of game-based learning in early childhood education. Game-based learning has a moderate to considerable impact on cognitive, social, emotional, motivational, and engagement outcomes, according to the research. According to the findings, game-based learning could be a helpful tool for early childhood educators looking to support young children's learning and growth. It will take more research to close the final gaps in the literatures.



According to Nokolova, This article presents a module for offering interactive educational resources to kids in internet and mobile environments. Because they lack the qualities necessary to satisfy the audience's expectations in terms of presentation, engagement, and focus, traditional e-learning platforms are inappropriate for toddlers. We suggest a collection of interactive software elements for developing and delivering family-friendly lessons in order to address this issue. Both new and old electronic learning systems and mobile applications can incorporate these elements. The essay highlights the target audience's needs, outlines the difficulties in utilizing learning management systems, and then discloses the features of a module designed to provide educational resources for children.

Today's children come from a generation that grew up with access to computers, tablets, and the internet. To create engaging learning experiences for these kids, it's critical to comprehend a few traits and abilities. These include the fact that they prefer non-linear information, are accustomed to fast gratification and copious amounts of information, enjoy networking, and multitasking.

The availability of educational apps for kids and their downloads have grown since the start of the COVID-19 pandemic. The study presents the findings of a comprehensive review of the literature regarding the age-rating schemes for digital content



(video and games) that are in use globally. It also exposes the shortcomings of the schemes that don't work well for selecting digital content for young audiences. Systems for grading content according to age are subjective and concentrate only on explicit material deemed dangerous for young children.

According to Behnam (2020), This study found that digital game-based learning (DGBL) has the potential to open new avenues for education, it is still unknown how DGBL applications may affect the creativity of young students. This study's primary goal is to find out if DGBL application technologies—tablets and smartphones—can help preschoolers (ages 3–6) become more creative. Therefore, over the course of the last ten years, the author has routinely reviewed the literature and found 67 articles, of which only 20 have been deemed to be more pertinent to the topic at hand. Based on the findings, the author analyzed studies on the features of DGBL technology using PRISMA's principles to improve learning and creativity. Four categories have been used for this classification: the goals of recent research, the effects of using technology (tablets).

According to Spirilla (2019), Though kid-focused educational apps are widely available, there is little proof of their ability to facilitate learning. This time, kids were required to pick up ten new terms to progress in a story-based mobile game that



requires kids to use their word knowledge. Study 1 employed a receptive vocabulary test in a lab-based between-subjects design with 4-year-olds from middle-SES families to determine whether the kids had picked up the vocabulary from the game. Compared to kids who didn't play the game, kids who did play it correctly answered more questions. As part of a bigger classroom intervention, low-SES preschoolers in Study 2 played the game four times using a within-subject design. On tests of their expressive and receptive vocabulary, kids demonstrated learning.

Educational games are a valuable tool for teaching vocabulary to young learners, especially those who are non-native speakers. According to Althagafi (2022), The study was conducted on forty Arab young students in a kindergarten in Saudi Arabia, focusing on their vocabulary acquisition. The study found that using educational games enhanced their ability to absorb and retain information, provided real-world applications, and increased learner engagement. The traditional method of repetition was used with the control group, while educational games were used in a more inventive way. This suggests that educational games can be a valuable tool for improving vocabulary instruction for young learners.

According to Derramas (2022), Based on their study the experiences of preschoolers playing computer games and the observations of their parents and teachers, It found that children



from extended nuclear families are more likely to be hooked on games. Parents view gaming as an earned privilege, while teachers suggest age-appropriate games should be allowed in moderation.

According to Lawrence (2021), Research on the impact of mobile device use on children's development is emerging, particularly in learning, language, and self-regulation skills. While interactivity benefits children's learning and self-regulation, naturalistic use may lead to poorer language and self-regulation. Pediatric clinicians can support families in managing mobile device use, while future research should explore the mechanisms underlying negative effects and how content and context influence these relationships.

The general goal of the study was to find out the impact of online educational videos on a four-year-old child's vocabulary acquisition, pronunciation, and comprehension. According to Mazo(2023), The research, which involved observation, assessment, video recording, and interviews, found that watching these videos positively influences vocabulary acquisition, pronunciation, and comprehension. It also helps develop intellectual, social, emotional, and spiritual aspects of the child. However, the study suggests that watching these videos under parental supervision is beneficial. The findings also



support the relevance of Piaget's assimilation and accommodation theory in today's technology-driven environment.

Local Literature

According to Alviz (2018), Research Paper (postgraduate) from the year 2018 in the subject Didactics - Common Didactics, Educational Objectives, Methods, language: English, abstract: The main objective of this study is to incorporate explicit instruction and GBL for early grades reading to improve reading fluency. The study also aims to probe if significant correlation exists between explicit instruction and GBL.

According to Bagtaso (2016), In the article they highlighted it came out to be that This study aims to create a mobile application for toddlers and preschoolers to enhance their learning in reading, spelling, and mathematics. Using the scrum model and Android technology, the researchers developed a game-based system using Programming White-box testing was conducted to ensure functionality. The results were highly effective, and the researchers recommend full implementation for future challenges and opportunities in the mobile application market.

Mobile applications created for educational purposes have increased significantly as a result of the widespread use of mobile computing devices according to Fernando (2021). Educators can



create inventive location-based mobile learning experiences by utilizing gamification strategies in conjunction with the location services offered by those devices. Our analysis of educational mobile location-based games takes place in this review of literature.

According to Papadakis (2018), The study examines the popularity of mobile apps for young children has led to debates about their suitability for education. A study examined whether Greek preschool apps were designed according to developmentally appropriate standards. Results showed most apps focused on basic numbers and letters, promoted rote learning, and failed to deepen conceptual understanding. The study highlights the need for more comprehensive educational apps.

According to Neumann (2018), Found that their study examines the impact of iPads and literacy apps on emergent literacy skills in English-speaking children aged 2-5 years found that tablets positively support letter name and sound learning and emergent writing development. The study also found no significant differences in letter writing skills or numeral knowledge. Further investigation is needed to determine the best use of these digital tools in early childhood classrooms.

This article looks at how toddler's relationships with media, while technology becomes more digital, portable, and accessible



to everyone, in particular, we look at laws intended to protect kids from harmful content and whether or not the help parents deal with the social media platform. There are numerous ways that internet offer chances for learning and interaction, despite worries about exposing kids to harmful information. There is proof of advantages like the development of literacy and problem-solving skills, even though more research is required to completely comprehend that using mobile media like apps results in different outcomes than using traditional media like television.

Vaala & Jordan (2017).

Toddlers now frequently play with interactive technologies: it is not uncommon for toddlers to replace touchscreen games for outdoor games. Here, we examined the direct impacts of touchscreen and physical play on the cognitive flexibility of 2.5-year-olds—a crucial component of executive function. Toddlers played physically or with touchscreens for nine minutes, while the third group (control group) colored and drew. Cognitive flexibility was then assessed using a sorting task. The group that engaged in physical play performed better than the other two. While touchscreen play had no impact on toddlers' cognitive flexibility, physical play improved it when compared to the control group.

Golden West Colleges Inc

College of Information Technology Education

performed better than those who viewed gaming as a solitary activity: Andrés & Wang (2016)

Synthesis and Research Gap

According to the study related to toddler's learning games, games have a moderate to large effect on cognitive, social, emotional, motivation, and engagement outcomes. Games can support children's sense of autonomy. However, there are still some research gaps in this study. First, too much gaming can lead to addiction, social isolation and additionally decreased academic performance.

This chapter has presented a comprehensive review of the literature related to toddler's learning games. The finding from these studies provides a foundation for the current research, which aims to have a positive impact on children's brain development and provide a fun and engaging way for children to enhance and having advanced learning skills. This study contributes to the understanding of learning games for toddlers.

CHAPTER 3

DESIGN & METHODOLOGY

Research Design

Quantitative data provides measurable information that can be used to make decisions, evaluate performance, and analyze trends. Quantitative research is a method of analyzing data that uses statistical techniques surveys are a common method for quantitative research because they can reach a large number of people and are easy to distribute.

The purpose of this chapter is to design the methodology of the research approach through quantitative research techniques. The research approach also supports the researcher on how to come across the research result findings. In this chapter, the general design of the research and the methods used for data collection are explained in detail. The purpose of this section is to indicate how the research was conducted throughout the study periods.

In the people lived in urban areas the developers conducted a survey and managed to asked more information needed from these clients to help the system to be good for toddlers.

Software Developmental Model

Rapid Application Development (RAD) is built on development and testing and an agile approach with little (or no)

preparation ahead of time. Taking a RAD approach to software development involves focusing less on preparation than on the development and advancement of a project. RAD is a development paradigm that arose from realizing that the conventional waterfall model of development was ineffective.

The waterfall prototype has a big drawback in that it's impossible to modify the software's key features and specifications until it's in the testing process. As a result, you're left with applications that can or cannot meet your changing needs. The waterfall prototype has a big drawback in that it's impossible to modify the software's key features and specifications until it's in the testing process. As a result, you're left with applications that can or cannot meet your changing needs. (Kissflow,2024).

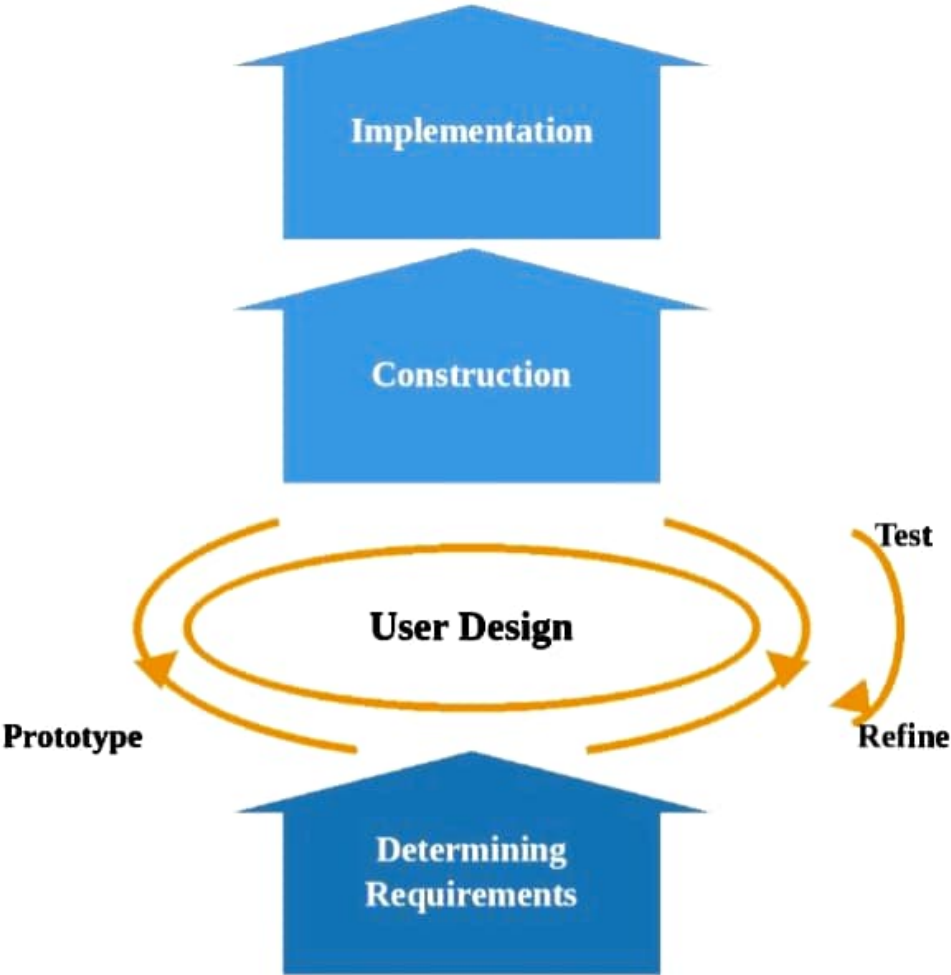


Figure 3.1 Rapid Development Cycle

The Stages of Rapid Application Development

Define the requirements: Rapid application development distinguishes itself from standard computing patterns right away. It doesn't ask for a comprehensive list of requirements from end-users; rather, it asks for a wide criterion. The general structure of the specifications makes it easier to include exact needs at various stages of the development process.

The developers will conduct and collect the required data from the client to include the necessary things. The agreement will be easier for the client and developers before going to the next stages.

Prototype: This is where much of the action takes place. Rather than attach to a specific set of specifications, companies create designs of various functionality and configurations as quickly as possible. The clients are then shown the samples and asked to determine what they do and don't like. Most of the time, these designs are rushed to function to demonstrate specific functionality without a sufficient finish. It is natural, and the finished result is only generated when both the customer and the programmer agree on the final product at the completion phase.

In this stage, the developers and the client will work together to remove and attach some unnecessary and needed for the system until the system is ready for client evaluation.

Receive feedback: Feedback on what's fine, what's not, what functions, and what doesn't is shared at this point. Not only is feedback restricted to pure features, but it also includes graphics and frameworks. Prototyping begins with this feedback in consideration. These two processes are repeated before a finished design is produced that meets the needs of both the programmers and the consumers. In this stage, the user/ client will test the system.

The developers will receive feedback from the client whether it is not or within the needed/required standard. The prototype also evolved, Finalize Software: The consumer finalizes the software's specifications, operations, layout, and interface here. Until providing to the customer, reliability, accessibility, and maintainability are critical. This is the last stage wherein the developers will finalize the documentation and completed the system before handling it to the user/client.

Source of Data

The developers have conducted information from the two sources.

First, having personal survey interview with the people lived in urban areas. The developers gained important thoughts and opinions from this people about benefits of having online learning games for toddlers using the survey we made for them

to answer, majority of them answered yes and the things that might help for their toddler's brain Development.

The majority of them said that toddler's learned basic concepts such as number, letters, colors, and sounds in advance and that's a very good sign for toddler's brain development.

The last and second source of data is by collecting information from the internet. The developers gained information like videos and articles from world wide web. All the collected information from these sources is carefully analyzed and is related to the proposed system, The Toddler's learning game.

Instrumentation and Data Collection

In this section is about data gathering by the developers, a research component to measure and analyze the information if it's accurate to the study.

Survey. Surveys are useful in describing the characteristic of a large population. No other research method can provide this broad capability, which ensures a more accurate sample to gather targeted results in which to draw conclusions and make important decisions.

The developers conducted a survey with the client to gained more information and recommendations. At first, the developers prepared 50 copies of survey questionnaires and conducted a survey to the people lived in Urban areas.

Online Research. Online research methods help people find the information they need quickly. Online tools provide information instantly. This information would take much longer to locate using offline research methods (such as searching for books in a library).

The developers used online research mostly to find information and collect more data from different articles related to the study.

Tools for Data Analysis

The developers used the following tools to collect data as guide for the proposed system. The collected data are deeply analyzed and understand the related data for developing the system called Toddler's learning games.

Database. A database is a structured, organized collection of data stored on a computer system. Databases typically store information in multiple linked tables that keep relevant data for each record in dedicated fields. Database management system (DBMS) software maintains the relationships between tables, adds and updates records, and displays data in response to queries by Brian P. (2023). Database developers used database to make sure that all the data are stored in one place.

Database schema. A database schema is a blueprint for the way your data is organized and structured. In other words,

this is a set of rules for what data can be stored, along with how and where. Note that this is separate from the actual values that your database will store. Producing schema means mapping out the values your database will store, how they'll relate to each other, and the rules that will govern them. To put it another way, a schema is a blank canvas for your data. Later, you can start to populate this with actual values. The goal is to build a framework that closely matches your requirements for your finished app. Then, you can use this as the foundation of your business process layer, and your user interface. (Ronan McQuillan 2022).

The database schema determines the storage rules for database. It also provides the framework for user interfaces and app requirements. Database schemas provide clear documentation, which helps developers to understand the database structure of our system.

Entity Relationships Diagram. An entity-relationship diagram (ERD) is crucial to creating a good database design. It is used as a high-level logical data model, which is useful in developing a conceptual design for databases. An entity is a real-world item or concept that exists on its own. Entities are equivalent to database tables in a relational database, with each row of the table representing an instance of that entity. An attribute of an entity is a certain resource that describes the entity. A relationship is the connection that describes the

interaction between entities. Cardinality, in the context of ERD, is the number of instances of one entity that can, or must, be related with each case of another entity. In general, there may be one-to-one, one-to-many, or many-to-many relationships. (Margaret Rouse 2017).

Is a crucial thing to our developers. But it's important to make a database design. To know what entity or design we represent.

Flowchart. A Flowchart is a visual representation of a process, system, or algorithm. Flowcharts use symbols like rectangles, diamonds, and arrows to depict the steps, decisions, and possible outcomes involved. Think of it like a roadmap that helps you understand the "big picture" and navigate the details of a process. (Pat Timmons 2024).

In the development stages, flowcharts improve understanding for audience and also essential tool for system creation. It helps developers arrange their thoughts and structure logic to reduce errors and improve code quality.

Gantt Chart. The Gantt Chart is a project management tool that displays the project schedule as a bar chart. It is used for planning and tracking the project's progress and is the most useful way to show activities against time. (Andrei T. 2024).

The Gantt Chart is used as time management tool. It is the most effective way to manage project plans on time and progress

tracking. The developers use the Gantt chart to split into smaller task to make it easier to manage the tasks.

Case Diagram. A use case diagram is a model of system functionalities as seen from the perspective of users and other external entities, which allows you to understand system requirements and its communication with various actors. (Marta Rutkowska 2023).

The developers allow to expand our system for fasting communicate to others.

CHAPTER 4
FINDINGS & DISCUSSION

Flow Chart

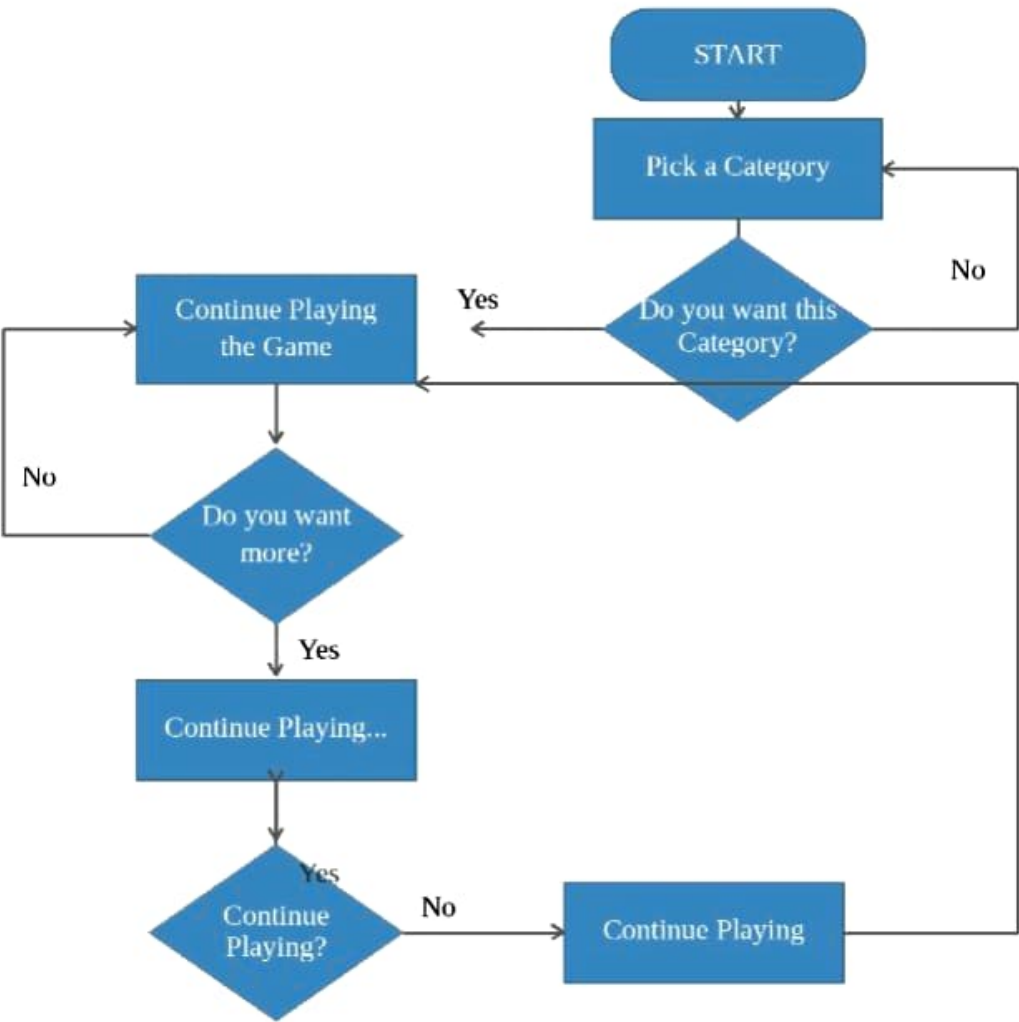


Figure 4.1 Flow Chart of the Game

Graphical Representation of Data

1. Are there children 3-6 years old in your household?

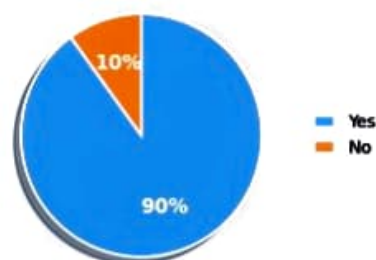


Figure 4.2 Survey Question No.1

The 90% of respondents have children 3-6 years old or accompanying children 3-6 years old in their home and 10% of them said No.

2. What is your relationship with the child?

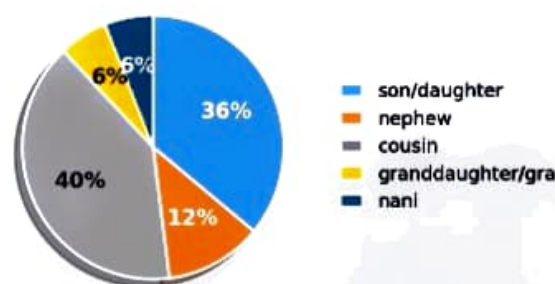


Figure 4.3 Survey Question No.2

Majority of the result, 40% of the respondents are related to the child/toddler as their cousin, 36% for son/daughter, 12%

for nephew, 6% for granddaughter/grandson, and the least is the Nani or Babysitter 6%.

3. Does your Toddler/s use gadgets from time to time?

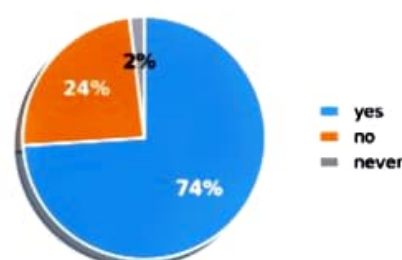


Figure 4.4 Survey Question No.3

The 74% of the respondents said that their toddler's (3-6 y/o) used gadgets from time to time. 24% of them said no and 2% of them said never.

4. How often do you participate in guiding your Toddler's using different devices such as mobile phones, tablet and ipad.

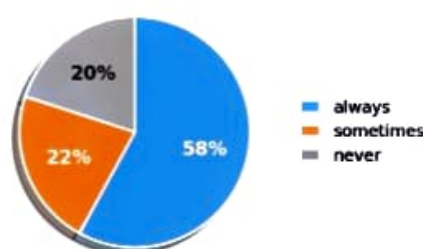


Figure 4.5 Survey Question No.4

According to the data, 58% of the respondents are focused and watching over their children while using their gadgets and 22% of them said sometimes and 20% said never.

5. Does your Toddler/s use gadgets from time to time?

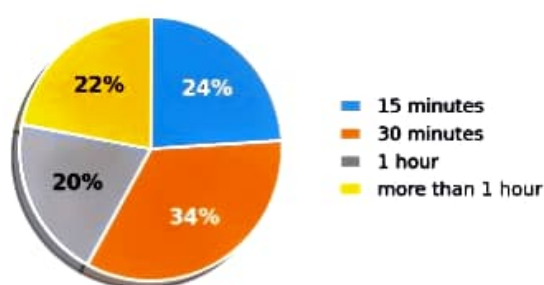


Figure 4.6 Survey Question No.5

Almost all respondents answered the survey equally. 34% for 30 minutes, 24% for 15 minutes, 20% for 1 hour, and 22% for more than an hour.

6. Do you think learning games can help toddler/s in their social development?

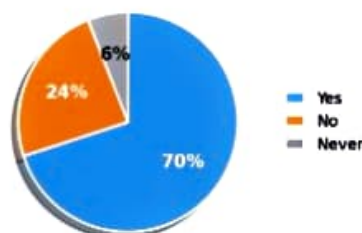


Figure 4.7 Survey Question No.6

Most respondents said yes, therefore 70% of them think that learning games can be helpful for their social development, but 24% of them said no and 6% for never.

7. Do you think learning games can help toddler/s in their emotional development?

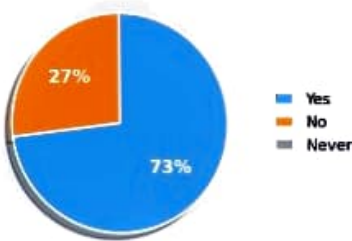


Figure 4.8 Survey Question No.7

Most respondents said yes, therefore 73% of them think that learning games can be helpful for their emotional development, 27% of them said no, and 0% said never.

8. Do you think learning games can help Toddler/s in their mental development?

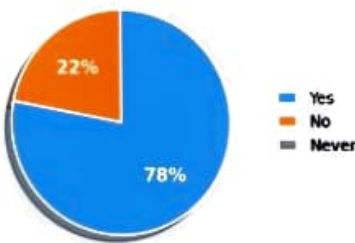


Figure 4.9 Survey Question No.8



Most respondents said yes, therefore 78% of them think that learning games can be helpful for their mental development, 22% of them said no, and 0% said never.

9. What are the possible benefits that Toddler/s can get while or after using toddler's learning game?

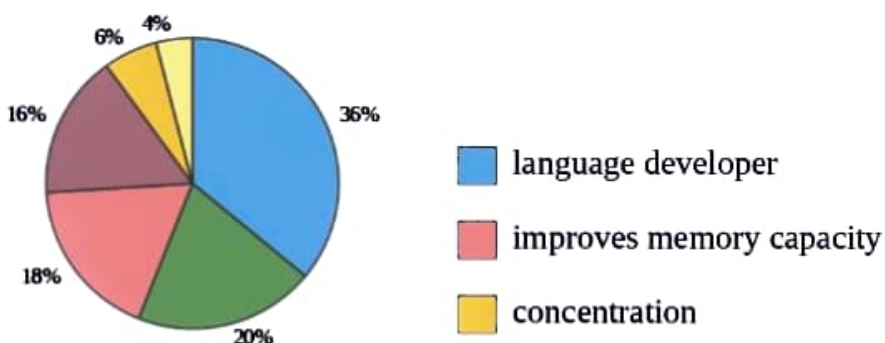


Figure 4.10 Survey Question No.9

The respondents have been made so many decisions but the one with the highest votes is the Language Development with 36%. 20% for advance learning before going to school, 18% for improves memory capacity, and 16% for creativity, concentration with 6% of votes the lowest votes is the Learning from mistakes with the total percent of 4%.



10. In your opinion, how effective are learning games in helping your Toddler/s with the following skills.

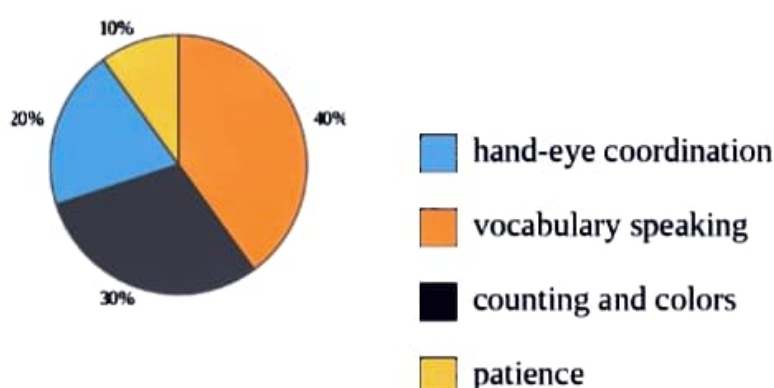


Figure 4.11 Survey Question No.10

According to the data, 40% of the respondents rate the highest is the vocabulary and speaking, second is the counting and colors with 30% of rate, 20% for hand-eye coordination and lastly 10% for patience.

Problem Areas

The problem that we found is that the children can use it an infinite amount of time and can cause them with blurry eye by playing the game too much.

It might be boring to some toddler's because of lack of creativity



Developed Game



Plates 4.1 Game Notice

This is where the notice for parents pops up.



Plates 4.2 Game Category

This is where the toddler's pick a category.



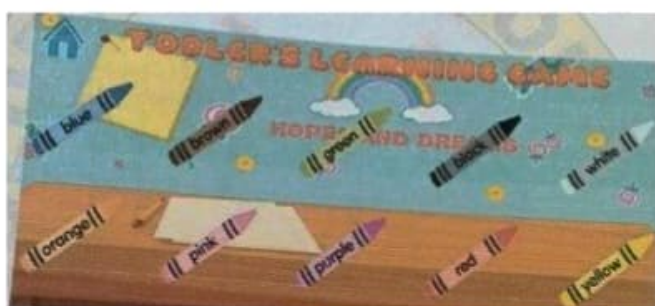
Plates 4.3 Animal Category

This where the toddler pick animals



Plates 4.4 Animal Pick

This where the toddler learn animals. They can either next or previous.



Plates 4.5 Color Category

This where the toddler pick color.



Plates 4.6 Color Pick



This where the toddler learn colors. They can either next or previous



Plates 4.7 Letter Category

This where the toddler pick a letter.



Plates 4.8 Letter Pick

This where the toddler learn letters. They can either next or previous.



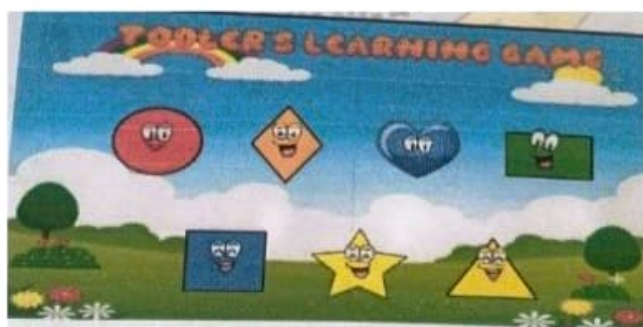
Plates 4.9 Number Category

This where the toddler pick number



Plates 4.10 Number Pick

This where the toddler learn numbers. They can either next or previous.



Plates 4.11 Shapes Category

This where the toddler pick shape.

CHAPTER 5

SUMMARY, CONCLUSION, & RECOMMENDATION

This chapter presents the summary of findings, conclusions and the recommendations made by the developers. Games and game-based learning are used because they have many benefits; especially, games and play are strongly connected to children's cognitive development and help develop many cross-curricular skills as a result.

This study was based on the results of the data gathered from the 50 respondents' questionnaire. An analysis of the content was done with the use of pie graph method to get the total answers of 50 respondents.

Summary of the Findings

The findings of the study were arranged according to statement of the problems stated in Chapter 1.

- 1) Toddler's learning games is a jump-start for toddler's brain development by engaging in letters, sounds, and numbers.
- 2) When the toddlers seem to have the ability to listen, understand the activities, and follow instructions then the toddler is ready.
- 3) Observed – Pay attention to how the toddlers interact with people. Have a conversation with the toddler to learn more.

about their knowledge and ensure that the activities are appropriate for the child.

- 4) Toddlers learn best when they're interested in an activity. Let them make mistakes until they know what is right. Teach them healthy ways to express their emotions.

Conclusions

In this study, the developers prepared 50 questionnaires for the target client. And using the Pie Graph method the developers got the total votes of the respondents and got the right answer that needs to be the outcome for the statement of the problem stated in chapter 1.

The findings of this research affirm that learning games significantly improves mental, social, and emotional well-being of the toddlers.

While this study provides valuable insights into the Toddler's and the Guardian, the developers need a warrant further knowledge about the possible physical effects for toddlers. The developers offer a maximum of 1 hours per day using the toddlers' learning games and emphasize the potential of physical activity as a preventive measure against addiction of using the Toddler's learning games.

Recommendations

In the light of the findings and conclusions, this are offered as recommendations for possible actions:

- 1) The child should not last an hour using gadgets;
- 2) Make sure to supervise the child while doing this activity.
- 3) Speech games for toddlers that allows them to gained knowledge to improved their speech and also develop proper grammar.
- 4) Can be used without using internet;
- 5) Mobiles used, can be used anywhere and anytime;
- 6) Different kind of animals like Amphibians, Mammals, Birds and many more; and
- 7) Counting numbers from 1 through 10.

BIBLIOGRAPHY

A

Alotaibi (2024). Game-based learning in early childhood education a systematic review and meta-analysis.
<https://doi.org/10.3389/fpsyg.2024.1307881>

Alviz 2018: Utilizing Game-based Learning in Explicit Instruction for Early Grades Reading Fluency.
https://www.google.com.ph/books/edition/Utilizing_Game

B

Bagtaso et al. (2016), Mobile Educational Games for Toddlers and Preschoolers Available at SSRN: <https://ssrn.com/abstract=3725174>

Bennani et al., (2020) The landscape of research on the use of digital game-based learning apps to nurture creativity among young children: Volume 37, September 2020.
<https://doi.org/0.1016/j.tsc.2020.100666>

C

Caneiro (2021) This article is an open access article distributed under the terms and conditions of the Creative Commons: Attribution Duc. Sci. 2021, 11, 229.
<https://doi.org/10.3390/educsci11050229>

Golden West Colleges
College of Information Technology Education

D

Derramas (2022). Preschoolers' Use of Computer Games: Basis for Parental and Educational Direction.

<https://ejournals.ph/article.php?id=20664>

H

<https://study.com/academy/lesson/educational-games-definition>

<https://www.bracknell-forest.gov.uk/children-and-family-services>

<https://ideascale.com/blog/what-is-research-design/>

<https://www.ilovephd.com/why-research-design-is-important-for>

<https://kissflow.com/application-development/rad/rapid-applica>

L

Lawrence et al., (2021). Mobile Media and Young Children's Cognitive Skills: A review.

Academic Pediatrics, 21(6), 996-1000.

<https://doi.org/10.1016/j.acap.2021.01.007>

M

Mazo (2023). Vocabulary Acquisition of a Four-Year-Old Child Through Piaget's Accommodation Theory.
<https://ejournals.ph/article.php?id=20665>

N

Neumann (2018). Using tablets and apps to enhance emergent literacy skills in young children. *Early Childhood Research Quarterly*, 42, 239-246.
<https://doi.org/10.1016/j.ecresq.2017.10.006>

Nikolova et al., (2020) Tools for Creating and Presenting Online Learning Resources for Preschool Kids.
[https://www.researchgate.net/publication/346580707_Too](https://www.researchgate.net/publication/346580707_Tools_for_Creating_and_Presenting_Online_Learning_Resources_for_Preschool_Kids)

P

Papadakis et al., 2017, Prior B and Renaud, 2020
International Journal of Child Computer Interaction Volume 33, September 2022,100480.
<https://doi.org/10.1016/j.ijcci.2022.100480>

Papadakis et al., (2018). Educational apps from the Android Google Play for Greek preschoolers: A systematic review. *Computers & Education*, 116, 139- 160.
<https://doi.org/10.1016/j.compedu.2017.09.007>

S

Saleh et al.,(2022)The Effect of Using Educational Games as a Tool in Teaching English Vocabulary to Arab Young Children: A Quasi-Experimental Study in a Kindergarten School in Saudi all authors and affiliations All Articles.

<https://doi.org/10.1177/21582440221079806>

Silva, et al. (2021) Literature Review of Location-Based Mobile Games in Education: Challenges, Impacts and Opportunities Informatics 2021, 8(3), 43;
<https://doi.org/10.3390/informatics8030043>

Spirilla et al. (2019): Education in the app store: using a mobile game to support U.S. preschoolers' vocabulary learning, Journal of Children and Media.
<https://doi.org/10.1080/17482798.2019.1650788>

Golden West Colleges
College of Information Technology Education



CORBILLON, AJ A.

BSIT STUDENT

About Me

I am AJ and I'm studying bachelor of science information Technology course. I take this course to enhance my skills and ideas through programming.

Contact

09104721075
ajcor492@gmail.com
Balingasay, Bolinao,
Pangasinan

Computer Skills

- WEB
- GRAPHICS
- SOFTWARE SKILLS

Language

- English
- Tagalog

☐ **Education**

ELEMENTARY SCHOOL

(2015 - 2016)

**LA HUERTA
ELEMENTARY
SCHOOL**

HIGH SCHOOL

(2018 - 2019)

**PARANAQUE
NATIONAL HIGH
SCHOOL**

SENIOR HIGH SCHOOL

(2020 - 2021)

**TECHNICAL
VOCATIONAL
LIVELIHOOD
COOKERY (TVL)**

☐ **References**

Mr. Kairos Jil Molina
Dean, College of
Information Technology
Golden West College

**Mr. John Vianney V.
Manuel**

SAD, Admission

APPENDICES

APPENDIX A

Survey Question

TODDLER'S LEARNING GAMES

Name: _____ (Optional) Date: _____

Guide Questions:

1. Are there children 3-6 years old in your household?
☐ Yes ☐ No
2. What is your relationship with the child?
☐ Son/Daughter
☐ Nephew
☐ Cousin
☐ Granddaughter/ Grandson
☐ Nani/Babysitter
3. Does your Toddler's use gadgets from time to time?
☐ Yes ☐ No ☐ Never
4. How often do you participate in or guide your Toddler's using different devices such as mobile phones, tablet and iPad?
☐ Always ☐ Sometimes ☐ Never
5. How long, on average, does your toddler spend playing learning games?
☐ 15 minutes ☐ 30 minutes
☐ 1 hour ☐ More than 1 hour
6. Do you think learning games can help Toddler's in their social development?
☐ Yes ☐ No ☐ Never
7. Do you think learning games can help Toddler's in their emotional development?
☐ Yes ☐ No ☐ Never
8. Do you think learning games can help Toddler's in their mental development?
☐ Yes ☐ No ☐ Never
9. What are the possible benefits that toddlers can get while or after using toddler's learning games? Give at least 2 benefits
☐ Language development
☐ Improves memory capacity
☐ concentration
☐ creativity
☐ Learning from mistakes
☐ advance learning before going to school
10. In your opinion, how effective are learning games in helping your Toddler's with the following skills. (Rate 1-10 ex. Patience- 8/10)
☐ hand-eye coordination -
☐ vocabulary, speaking -32
☐ counting and colors -
☐ patience -

APPENDIX B

Sample Survey

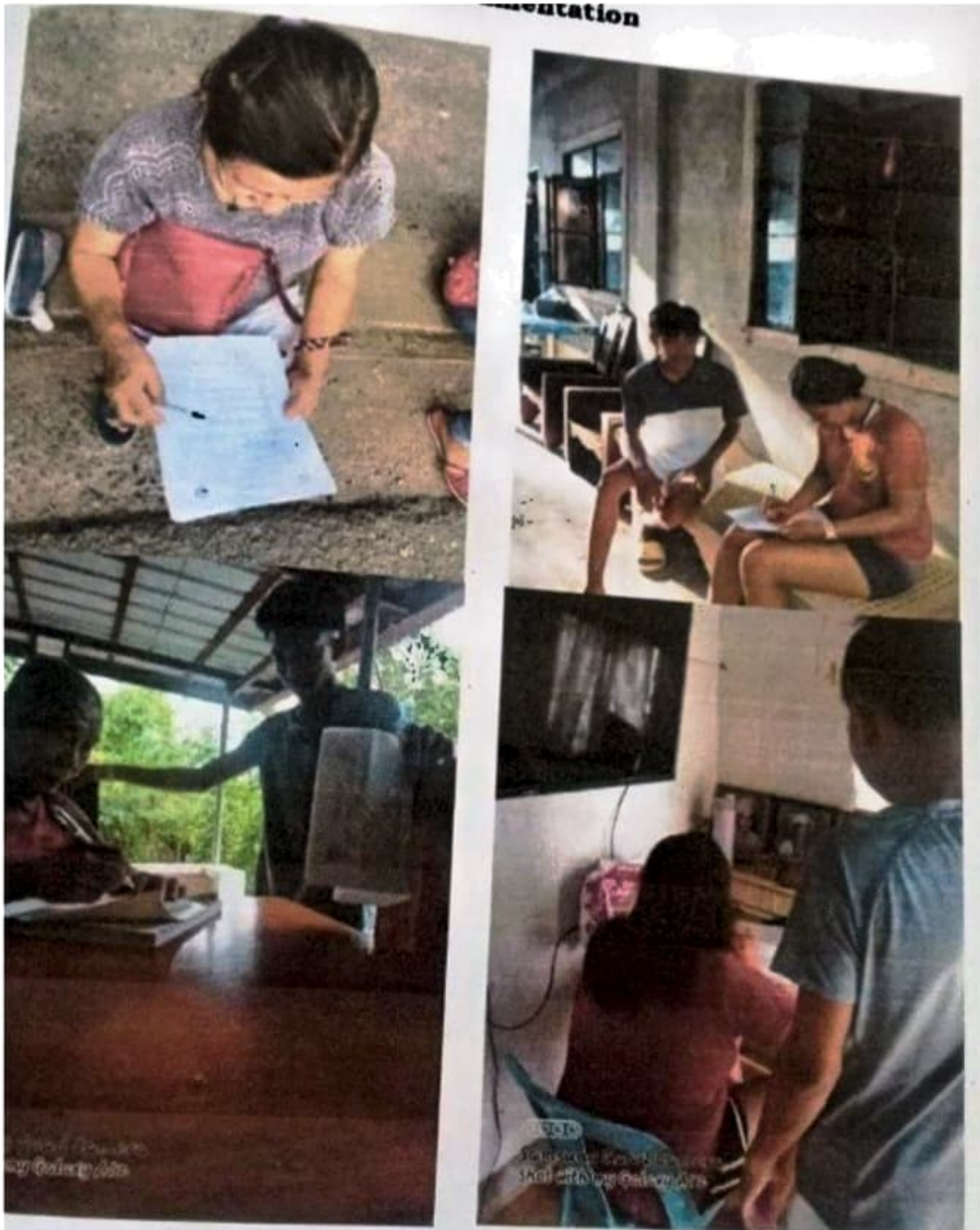
TODDLER'S LEARNING GAMES

Name: _____ Age: 3–5 years old (Optional) Date: _____

Guide Questions:

- 1. Are there children 3 – 5 years old in your household?**
 - a. Yes
 - b. No
- 2. What is your relationship with the child?**
 - a. Son/Daughter
 - b. Nephew
 - c. Cousin
 - d. Granddaughter / Grandson
 - e. Niece/Nephew
- 3. Does your Toddler's use gadgets from time to time?**
 - a. Yes
 - b. No
 - c. Never
- 4. How often do you participate in or guide your Toddler's using different devices such as mobile phones, tablet and iPad?**
 - a. Always
 - b. Sometimes
 - c. Never
- 5. How long, on average, does your toddler spend playing learning games?**
 - a. 15 minutes
 - b. 30 minutes
 - c. 1 hour
 - d. More than 1 hour
- 6. Do you think learning games can help Toddler's in their social development?**
 - a. Yes
 - b. No
 - c. Never
- 7. Do you think learning games can help Toddler's in their emotional development?**
 - a. Yes
 - b. No
 - c. Never
- 8. Do you think learning games can help Toddler's in their mental development?**
 - a. Yes
 - b. No
 - c. Never
- 9. What are the possible benefits that toddlers can get while or after using toddler's learning games? Give at least 2 benefits.**
 - a. Language development
 - b. Improves memory capacity
 - c. Concentration
 - d. Creativity
 - e. Learning from mistakes
 - f. Advance learning before going to school
- 10. In your opinion, how effective are learning games in helping your Toddler's with the following skills? (Rate 1–10, ex. Patience – 8/10)**
 - a. Eye-hand coordination – 5/10
 - b. Vocabulary speaking – _____
 - c. Naming and colors – _____
 - d. Patience – _____

APPENDIX C
Survey Documentation



APPENDIX D
Source Code



https://drive.google.com/drive/folders/1qfTbGOf5LKWDv0PyQ83qprbD5MXkYGH?usp=drive_link

APPENDIX E
User Manual

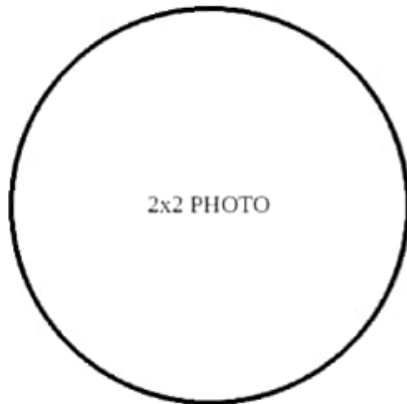
How to install the game

1. Scan the QR code below.
2. Open the link.
3. Download the Game.
4. Install the game then enjoy.

How to operate the game

1. Open the game.
2. Agree in the notice.
3. Play the game and enjoy.

CURRICULUM VITAE



**GANADEN,
CHERRY ANN D.**
BSIT STUDENT

Education

ELEMENTARY SCHOOL
(2015 – 2016)
SAMPALOC BOLINAO ELEMENTARY SCHOOL

HIGH SCHOOL
(2019 – 2020)
BANI NATIONAL HIGH SCHOOL

SENIOR HIGH SCHOOL
(2021 – 2022)
Information and Communication Technology (ICT)
SAMPALOC BOLINAO SENIOR HIGH SCHOOL

References

Mr. Kairos Jill Molina
Dean, College of Information Technology
Golden West College
Mr. John Vianney V. Manuel
SAD Advisor
Golden West College

About Me

I am Cherry Ann D. Ganaden, and I'm studying an Information Technology course, which I think is timely and related to the strand I took when I was in senior high.

Contact

09*****0
dche8277@gmail.com
Sampaloc, Bolinao, Pangasinan

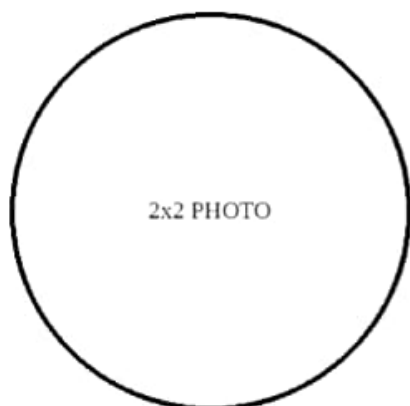
Skills

- Willing to learn
- Accurate
- Hard working

Language

- English
- Tagalog

CURRICULUM VITAE



**BALLARES,
MARKSHANE**
BSIT STUDENT

Education

ELEMENTARY SCHOOL
(2015 – 2016)
MABINI CENTRAL SCHOOL

HIGH SCHOOL
(2019 – 2020)
TAGUDIN NATIONAL HIGH SCHOOL

SENIOR HIGH SCHOOL
(2020 – 2021)
Technical Vocational Educational Agri-Fisheries (TVL)
TAGUDIN NATIONAL HIGH SCHOOL

References

Mr. Kairos Jill Molina
Dean, College of Information Technology
Golden West College
Mr. John Vianney V. Manuel
SAD Advisor
Golden West College

About Me

I am Mark Shane Ballares, an IT Student in Golden West College.

Contact

09306573522
markshaneballares02@gmail.com
Pob. Mabini, Pangasinan

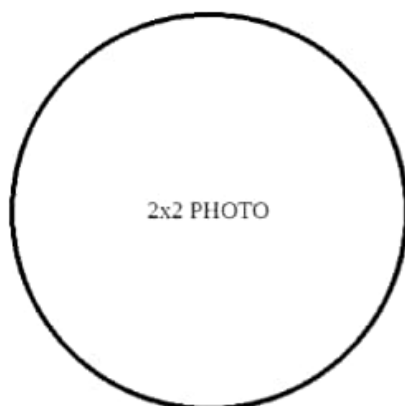
Skills

- Always ready
- Hardworking
- On time

Language

- English
- Tagalog

CURRICULUM VITAE



**BALLADO,
ROGELIO**
BSIT STUDENT

Education

ELEMENTARY SCHOOL
(2013 – 2014)
CATO ELEMENTARY SCHOOL

HIGH SCHOOL
(2018 – 2019)
CATO NATIONAL HIGH SCHOOL

SENIOR HIGH SCHOOL
(2020 – 2021)
Technical Vocational Educational Agri-Fisheries (TVL)
CATO NATIONAL HIGH SCHOOL

References

Mr. Kairos Jill Molina
Dean, College of Information Technology
Golden West College
Mr. John Vianney V. Manuel
SAD Advisor
Golden West College

About Me

I am one who is always excited to learn new things and accomplish given tasks. I am also someone who puts my peers before myself and helps them whenever required.

Contact

09812897544
bossjet24@gmail.com
Cato, Infanta, Pangasinan

Skills

- Strong organizational skills and ability to prioritize work
- Quick to learn and adapt to new environments and technologies
- Efficient at managing time and meeting deadlines

Language

- English
- Tagalog